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Bridging the Scale Gap: Balanced Tiny and General Object Detection in Remote Sensing Imagery

Zhicheng Zhao, Yin Huang, Lingma Sun, Chenglong Li, Jin Tang

***Abstract*—Tiny object detection in remote sensing imagery has attracted significant research interest in recent years. Despite recent progress, achieving balanced detection performance across diverse object scales remains a formidable challenge, particularly in scenarios where dense tiny objects and large objects coexist. Although large foundation models have revolutionized general vision tasks, their application to tiny object detection remains unexplored due to the extreme scale variation and density distribution inherent to remote sensing imagery. To bridge this scale gap, we propose ScaleBridge-Det, a large-scale detection framework specifically designed for tiny-object detection in remote sensing imagery, which could achieve balanced performance across diverse scales through scale-adaptive expert routing and density-guided query allocation. Specifically, we introduce a Routing-Enhanced Mixture Attention (REM) module that dynamically selects and fuses scale-specific expert features via adaptive routing to address the tendency of standard MoE models to favor dominant scales. REM generates complementary and discriminative multi-scale representations suitable for both tiny and large objects. Furthermore, we present a Density-Guided Dynamic Query (DGQ) module that predicts object density to adaptively adjust query positions and numbers, enabling efficient resource allocation for objects of varying scales. The proposed framework allows ScaleBridge-Det to simultaneously optimize performance for both dense tiny and general objects. Extensive experiments on benchmark and cross-domain datasets demonstrate that ScaleBridge-Det achieves state-of-the-art performance on AI-TOD-V2 and DTOD, while exhibiting superior cross-domain robustness on VisDrone.**

***Index Terms*—Aerial image, Tiny object detection, Scale balance.**

I. INTRODUCTION

REMOTE sensing object detection is a fundamental task in computer vision and serves as a critical component in various civilian applications, including environmental monitoring, urban construction, disaster response, and surveillance [1], [2], [3], [4]. The rapid development of unmanned aerial vehicle (UAV) and satellite imaging technologies has created a pressing need to automatically detect objects across diverse scales. Remote sensing images naturally exhibit large scale variations, spanning from tiny objects such as small vehicles or pedestrians, which occupy only a few pixels, to large structures like buildings and bridges that cover hundreds of pixels [5], [6] (depicted in Fig. 1). Furthermore, multi-scale targets commonly coexist within the same scene, posing significant challenges to current detection systems.

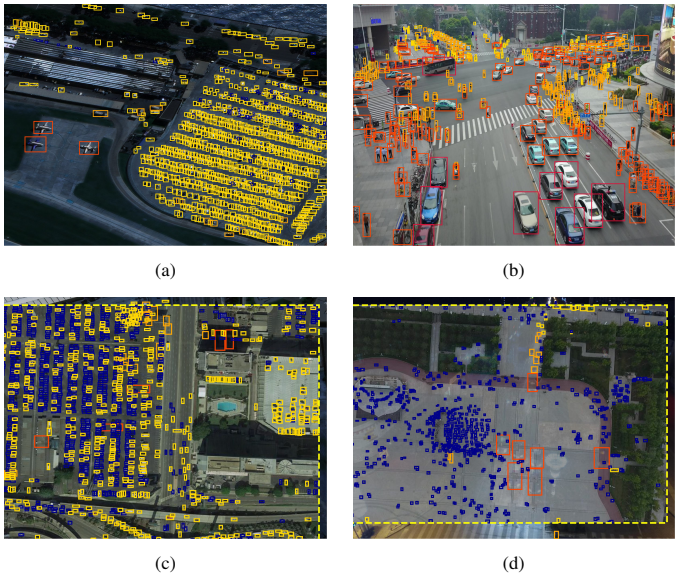


Fig. 1: Multi-scale object detection challenges in remote sensing imagery. (a)-(b) Representative scenes showing the coexistence of dense tiny objects and sparse large targets. (c)-(d) Visualization of extreme scale imbalance, with dense tiny object clusters (yellow dashed regions) juxtaposed with large structures. Colors indicate scales: dark blue (very tiny $< 8^2$), yellow (tiny 8^2-16^2), orange (small 16^2-32^2), orange-red (medium 32^2-96^2), crimson (large $> 96^2$).

Although object detection has achieved notable progress, balancing performance across diverse scales remains an open question (shown in Fig. 2), especially in remote sensing imagery [7], [8]. Context-enhanced detectors such as CFENet [9] strengthen tiny object features but lack global semantic reasoning. Transformer-based detectors (DETR [10], DINO [11], CoDETR [12]) capture global dependencies but exhibit notable performance imbalance for tiny objects [13], [14]. Query-based methods (DQ-DETR [14], D3Q [15]) adjust allocation based on density cues but do not address the underlying scale-completeness problem. Large detection models such as DINOv3 [16] demonstrate strong general detection through

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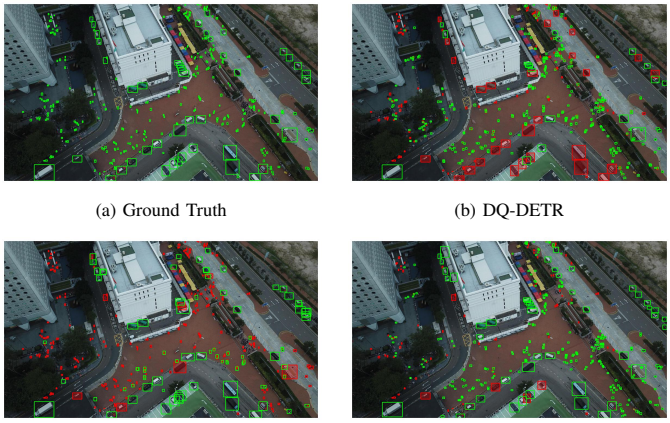
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(a) Ground Truth (b) DQ-DETR

(c) Ground-DINO (d) ScaleBridge-Det (Ours)

Fig. 2: Scale balance comparative analysis with maxdet=300 constraint on a representative scene containing 333 objects across five scale categories. Red boxes indicate missed detections. (a) Ground truth reference. (b) DQ-DETR excels at tiny objects but misses numerous general-scale targets. (c) Ground-DINO achieves strong general-object performance but fails on abundant tiny objects. (d) Our ScaleBridge-Det achieves balanced performance across all scales with minimal missed detections.

massive-scale training, yet lack explicit mechanisms to handle the extreme scale variations in remote sensing imagery [17], [18]. Moreover, most existing approaches rely on static feature fusion and uniform query allocation, which cannot satisfy the heterogeneous feature requirements across scales [19], [20].

To address this, we propose ScaleBridge-Det, a unified large framework that achieves balanced performance across diverse object scales through scale-adaptive expert routing and density-guided query allocation. Specifically, we construct a Routing-Enhanced Mixture Attention (REM) module that dynamically routes representations to the most suitable experts (ResNet, ViT, Swin Transformer), providing robust multi-scale representations for both tiny and general objects simultaneously. Furthermore, we propose a Density-Guided Dynamic Query (DGQ) module that adaptively adjusts query positions and numbers based on predicted object density, preventing tiny objects from being overwhelmed by the representational dominance of large targets. The proposed framework also demonstrates strong cross-domain robustness, generalizing well to different types of remote sensing imagery without extensive domain-specific fine-tuning.

The main contributions of this work are as follows:

We propose ScaleBridge-Det, a large detection framework tailored for tiny objects that leverages a mixture-of-experts architecture and progressive training to achieve balanced performance across extreme scale variations in remote sensing imagery. This addresses the fundamental scale competition problem where tiny objects are suppressed by dominant large structures.

We develop a Routing-Enhanced Mixture Attention (REM) module that dynamically selects and fuses scale-specific expert features through adaptive routing mechanisms. This generates robust multi-scale representations that prevent performance trade-offs between tiny and general object detection.

We propose a Density-Guided Dynamic Query (DGQ) module that predicts object density and adaptively adjusts query positions and numbers. This improves APvt and APt by 1.8% through scale-aware resource allocation and mitigates out-of-memory issues in dense scenes.

Extensive experiments demonstrate that ScaleBridge-Det achieves state-of-the-art performance with 35.7% mAP on AI-TODv2, 29.8% mAP50 on DTOD and 30.7% mAP on VisDrone under cross-domain evaluation with category mapping. It maintains balanced detection accuracy across all object scales while exhibiting superior cross-domain robustness as an additional benefit of the large model capacity.

II. Related Work

This section reviews the literature most relevant to this paper, organized into three categories: general object detection, tiny object detection methods, and large detection models.

A. General Object Detection

OBJECT detection in remote sensing imagery faces fundamental challenges that distinguish it from general computer vision tasks. Traditional CNN-based detectors such as Faster R-CNN [21] and RetinaNet [22] achieve strong performance on natural images but struggle when confronted with the extreme scale variations and dense distributions characteristic of aerial imagery [5], [23]. These methods fail to address how detectors can maintain balanced performance when tiny objects occupying fewer than 32×32 pixels coexist with large structures spanning hundreds of pixels within the same scene.

Transformer-based architectures have emerged as a foundational approach for capturing long-range dependencies in complex visual scenes. DETR [10] reformulates detection as a direct set prediction problem, eliminating hand-crafted components through end-to-end training. However, the method exhibits slow convergence and limited performance on small objects, achieving only 42% AP on COCO. Deformable DETR [24] improves convergence through deformable attention mechanisms that sample features at sparse key locations, yet the fundamental limitation persists. It still struggles to efficiently allocate attention across objects with extreme scale variations while maintaining balanced performance across all object scales.

DINO [11] advances DETR-like models by developing contrastive denoising training and mixed query selection, achieving 51.3% AP on COCO. CoDETR [12] proposes collaborative hybrid assignment training to enhance encoder-decoder optimization, reaching 64.4% AP. Despite these improvements, these methods face critical challenges in dense remote sensing scenarios [13], [14]. The core problem is that existing architectures lack mechanisms to prevent large objects from suppressing tiny object detection during training and inference.

Recent DETR variants, including DAB-DETR [25], DN-DETR [26], and Conditional DETR [27], develop dynamic anchors and conditional queries to improve convergence. Group DETR [28] introduces one-to-many assignment strategies for training efficiency. While these methods demonstrate potential

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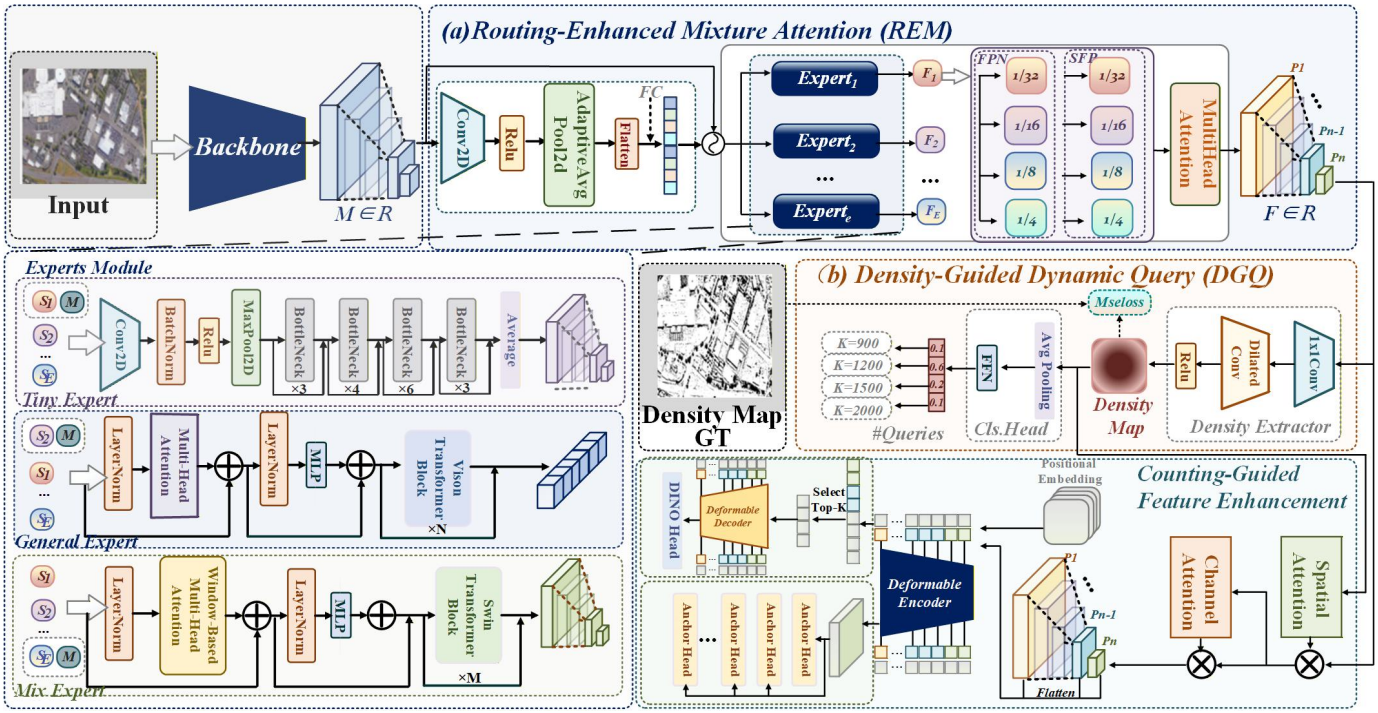


Fig. 3: Overview of the proposed ScaleBridge-Det framework, consisting of Multi-Expert Feature Extraction with adaptive routing, Routing-Enhanced Mixture Attention (REM) for scale-adaptive feature fusion, Density-Guided Dynamic Query (DGQ) for density-aware query allocation, and DETR Decoder for prediction refinement.

for capturing global context, they require substantial adaptation for remote sensing applications where the scale gap between tiny and large objects creates fundamental optimization conflicts [29], [30].

B. Tiny Object Detection

Tiny object detection in remote sensing presents distinctive challenges due to limited pixel representation, complex background clutter, and heterogeneous object densities [5], [23]. The core difficulty lies in extracting sufficiently discriminative features from objects occupying minimal spatial extent while maintaining computational efficiency, a constraint that fundamentally distinguishes this task from general object detection.

Context-enhanced approaches attempt to address feature representation challenges. CFENet [9] develops global and local context enhancement modules to improve small object features, achieving 27.8% APt on AI-TOD. However, overemphasis on local details leads to insufficient global semantic information, reducing APm performance by 4.2% [9]. This reveals a critical trade-off that methods optimized for tiny objects often sacrifice performance on larger ones. FPN-based methods [31] employ multi-scale feature pyramids to handle scale variations, but static allocation strategies fail when target densities vary dramatically across scenes [19]. MSNet [32] introduces local and global information fusion modules to enhance multi-scale feature representation, achieving strong performance on remote sensing benchmarks through dense connections across feature pyramid levels.

Feature enhancement techniques have been extensively explored. QueryDet [33] proposes adaptive query refinement for small objects, while TPH-YOLOv5 [34] develops transformer prediction heads to capture long-range dependencies in UAV images. DNTR [35] employs denoising feature pyramids to filter background noise, achieving improvements in APt and APvt on aerial datasets. TBNet [36] introduces texture and boundary-aware enhancement to strengthen discriminative features for small and weak objects through task-decoupled detection heads. Scale-aware attention networks [37] dynamically adjust receptive fields based on object scale, and SAHI [38] performs detection on sliced images to increase effective resolution. Despite these advances, these methods address symptoms rather than the root cause of scale imbalance, lacking mechanisms to explicitly balance performance across all object scales.

Density-aware methods tackle the problem of resource allocation in varying density scenarios. Fixed query mechanisms in DETR-like methods create inefficiency when object density varies dramatically—from sparse scenes with few objects to dense scenarios with hundreds of tiny targets [8]. DQ-DETR [14] develops categorical counting modules and density-guided feature enhancement, achieving 30.5% mAP on AI-TODv2 with 42.1% improvement in APvt for dense images. D3Q [15] refines this approach by explicitly estimating object density to determine optimal query numbers, reaching 32.1% mAP on AI-TODv2. DHLA [39] proposes dynamic hybrid label assignment to address the trade-off between one-to-one and one-to-many matching strategies in end-to-end detectors, improving representation learning through adaptive sample selection. Other approaches including Dense Query DETR [40], Adaptive Query Selection [41], and QueryProp [42] explore dynamic query strategies.

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Despite progress, existing approaches restrict their scope to density-based query adjustment while neglecting heterogeneous feature requirements across different scales [6], [20].

C. Large Detection Models

Large vision foundation models have achieved remarkable success through self-supervised learning at scale. DINOv3 [16] demonstrates strong COCO performance with frozen backbones, but general-domain models struggle with cross-domain generalization to remote sensing scenarios [17]. Domain adaptation techniques have been developed to address this challenge: Dual Teacher [43] proposes semi-supervised co-training for cross-domain ship detection, DA2-Net [44] integrates SAM2 with domain adaptation for remote sensing change detection, and CDTrack [45] develops multiple source domain adaptation for object tracking in satellite videos. LAE-DINO [46] introduces the first large-scale open-vocabulary detector for remote sensing with Dynamic Vocabulary Construction and Visual-Guided Text Prompt Learning. FAMHE-Net [17] further introduces multi-scale feature augmentation with heterogeneous experts.

Mixture-of-Experts (MoE) architectures activate subsets of parameters based on input characteristics, enabling large models with manageable computational costs [47], [18]. However, applying large models to tiny object detection in remote sensing remains under-explored. Existing routing strategies do not explicitly consider spatial feature variations across remote sensing modalities [6], [48]. This gap motivates our ScaleBridge-Det framework, which integrates scale-adaptive routing with density-aware mechanisms for balanced tiny-to-large object detection.

III. Method

This section first describes the framework of the proposed ScaleBridge-Det. Subsequently, the REM and DGQ modules are elaborated. Finally, the training strategy is introduced.

A. Overall Framework

The overall framework of the proposed ScaleBridge-Det is depicted in Fig. 3, which builds upon the CoDETR baseline [12]. ScaleBridge-Det consists of four key components: a backbone network, a REM module, a DGQ module, and a DETR decoder. Algorithm 1 presents the detailed pseudo-code of the complete ScaleBridge-Det framework. The algorithm takes an image tensor $\mathbf{I} \in \mathbb{R}^{B\times C\times H\times W}$ as input, where B is the batch size, C is the number of channels (typically 3 for RGB), H is the height, and W is the width. $\mathbf{I}$ is fed into the backbone for feature extraction. Furthermore, REM employs diverse backbone architectures, including ResNet, ViT, and Swin Transformer, to capture scale-specific representations. Scale-adaptive feature fusion is adopted to dynamically combine expert features through hybrid attention mechanisms. Then, the DGQ module adaptively adjusts query allocation based on predicted object density, incorporating a counting-guided feature enhancement mechanism that utilizes channel and spatial attention (CBAM) to refine encoder features according to density predictions, enabling more accurate query positioning in dense regions. Finally, the DETR decoder inherited from the baseline applies multi-layer self- and cross-attention, culminating in prediction heads for final bounding box regression and classification.

B. Routing-Enhanced Mixture Attention (REM) Module

The Routing-Enhanced Mixture Attention (REM) module addresses the fundamental challenge of integrating diverse expert features for multi-scale object detection. This module tightly couples routing-based expert selection with hybrid attention mechanisms for multi-scale feature fusion, ensuring that dynamic expert routing directly informs feature pyramid construction and enhances the model's capacity to handle diverse scales and feature characteristics [31]. We adopt ResNet-50, ViT-L, and Swin-L due to their complementary inductive biases. Specifically, ResNet-50 leverages local receptive fields that are effective for modeling tiny objects, ViT-L captures long-range dependencies through global self-attention, while Swin-L provides a balanced multi-scale representation via hierarchical window-based attention.

Adaptive Expert Routing. As illustrated in Fig. 3(a), the REM module employs an input-dependent dynamic routing mechanism that adaptively selects specialized experts based on image-specific characteristics [49], [50]. Given feature representations M from the base backbone (denoted as "$M \in R$" in Fig. 3), the routing network processes features through Conv2d, ReLU, AdaptiveAvgPool2d, Flatten, and FC layers to compute expert routing scores:

$$[s_1, s_2, \ldots, s_E] = \text{Softmax}(\text{FC}(\text{Flatten}(\text{GAP}(M)))) \quad (1)$$

where $s_e \in [0, 1]$ denotes the routing weight for expert e, indicating its importance for the current input.

To enhance computational efficiency and prevent feature dilution from low-contribution experts, we employ a selective activation mechanism that ensures at least one expert from each category is activated:

$$\mathcal{A} = \{e \mid e \in \text{Top-1}(\mathcal{C}_e)\} \quad (2)$$

where $\mathcal{A}$ denotes the set of activated experts, and $\mathcal{C}_e$ represents the category of expert e (tiny/general/mix). This category-wise top-1 selection guarantees diversity across expert types while maintaining computational efficiency. As shown in the "Experts Module" of Fig. 3, the activated heterogeneous experts produce feature outputs F_e for $e \in \mathcal{A}$, which are then weighted by the corresponding routing scores s_e for adaptive fusion.

Scale-Adaptive Feature Fusion. Following expert routing, each activated expert $e \in \mathcal{A}$ extracts multi-scale feature pyramids through FPN or SFP, producing features F_e^l at each pyramid level l.

To address the magnitude imbalance across heterogeneous expert architectures, we introduce a scale-aware normalization mechanism:

$$\widetilde{F}_e^l = \alpha_e \cdot F_e^l, \quad \forall e \in \mathcal{A} \quad (3)$$

where α_e is the learnable expert-specific calibration factor optimized end-to-end during training.

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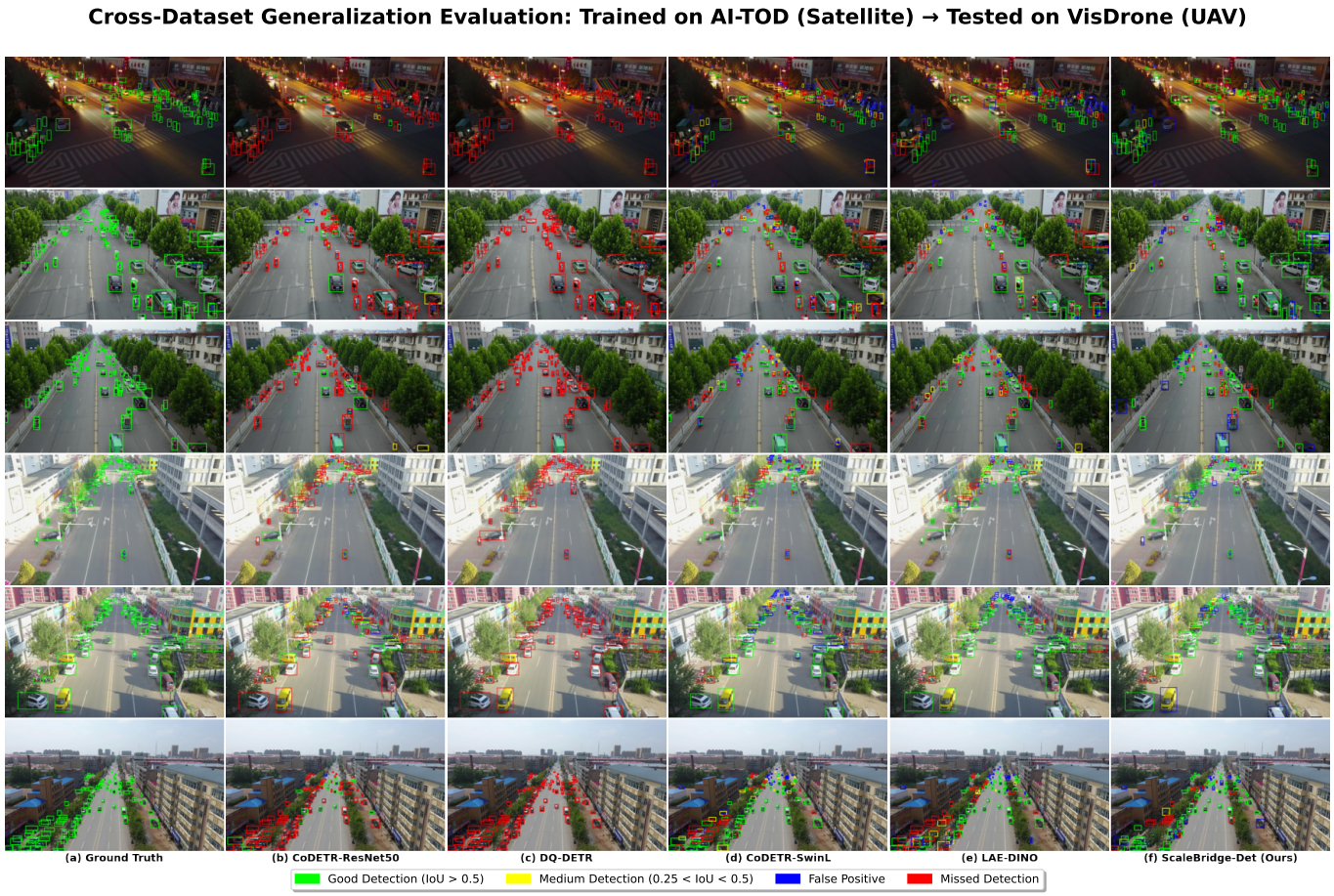


Fig. 4: Cross-domain generalization: trained on AI-TOD and tested on VisDrone without fine-tuning. Columns: (a) Ground Truth, (b) CoDETR-ResNet50, (c) DQ-DETR, (d) CoDETR-SwinL, (e) LAE-DINO, (f) ScaleBridge-Det (Ours). Green: IoU $>$ 0.5; Yellow: 0.25 $<$ IoU $<$ 0.5; Blue: false positives; Red: missed.

Building upon normalized features, a learnable dynamic gating mechanism computes scale-specific fusion weights. At each pyramid level l, the gating network aggregates expert features through concatenation and global average pooling, then computes level-specific gating scores:

$$g_l = \text{GAP}(\text{Concat}(\{\widetilde{F}_e^l \mid e \in \mathcal{A}\})) \tag{4}$$

$$w_l = \text{Softmax}(W_l \cdot g_l + b_l) \tag{5}$$

where W_l and b_l are scale-specific learnable parameters. The final fused representation at each scale combines both routing scores s_e and level-specific weights $w_{l,e}$ for activated experts only:

$$F_l^{\text{fused}} = \sum_{e \in \mathcal{A}} s_e \cdot w_{l,e} \cdot \widetilde{F}_e^l \tag{6}$$

The fused multi-scale features are subsequently processed through the Multi-Head Attention mechanism (shown in Fig. 3), which performs hierarchical feature integration using self-attention:

$$F^{\text{final}} = \text{MultiHead}(F^{\text{fused}}) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{7}$$

where Q, K, V are linear projections of the fused features. This generates the output feature pyramids $\{P_1, \ldots, P_n\}$ that serve as inputs to the downstream DETR decoder, balancing local details and global context across scales.

C. Density-Guided Dynamic Query (DGQ) Module

The Density-Guided Dynamic Query (DGQ) module enhances the detection of dense tiny object regions by providing spatial priors on object distribution [15]. Integrated with the highest-resolution feature map (256×256) from REM, the DGQ module combines density map generation with instance count-aware dynamic query allocation, significantly improving detection performance in high-density scenarios. Traditional fixed-query mechanisms prove insufficient for tiny object detection in remote sensing, where objects often cluster densely with limited pixel information [14]. The density map provides spatial priors that mitigate query starvation in dense regions and improve recall for small-scale targets [15]. Furthermore, in images containing thousands of targets, fixed-query approaches encounter out-of-memory (OOM) errors during Hungarian matching due to the quadratic complexity of the query count. The proposed dynamic adjustment mechanism scales queries appropriately to maintain both efficiency and performance. The detailed algorithm pipeline is presented in Algorithm 1.

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Density Map Generation. The ground truth density map is generated by convolving object centers with a Gaussian kernel, capturing local object densities. For each object instance centered at position $c_i = (x_i, y_i)$, a 3×3 Gaussian kernel with standard deviation $\sigma = 1.5$ (empirically optimized for tiny objects) is applied to produce the density map D^{gt}:

$$D^{gt}(p) = \sum_i \exp\left(-\frac{\|p - c_i\|^2}{2\sigma^2}\right), \tag{8}$$

where p denotes pixel positions, and the map is normalized to reflect relative densities. During training, a lightweight convolutional head (comprising 3×3 convolutions followed by 1×1 convolution) processes the high-resolution feature map to predict the density map D^{pred}. The module is supervised using mean squared error (MSE) loss for density map consistency, combined with auxiliary classification loss for semantic alignment:

$$\mathcal{L}_{density} = \|D^{pred} - D^{gt}\|_2^2 + \lambda_{cls}\mathcal{L}_{cls}(D^{pred}), \tag{9}$$

where $\mathcal{L}_{cls}$ is the binary cross-entropy loss encouraging high density predictions in object regions. This dual supervision ensures the predicted density map accurately approximates the ground truth distribution while effectively discriminating foreground from background [15].

The predicted density map serves as a spatial prior, highlighting regions with high object concentration. This approach draws inspiration from crowd counting techniques adapted for detection tasks, where density maps improve localization in cluttered scenes by providing coarse estimates of object counts and positions before fine-grained detection [24]. For tiny objects in aerial images, the density map compensates for detailed feature loss during downsampling in deeper layers, preserving cluster information that might otherwise be diluted.

Dynamic Query Selection. To adapt to varying object densities, the total instance count N is estimated by integrating over the predicted density map:

$$N = \sum D^{pred}, \tag{10}$$

rounded to the nearest integer. Based on this estimate, query numbers are dynamically adjusted across four tiers to balance computational efficiency and detection capacity:

$$\text{Queries} = \begin{cases} 900 & \text{if } N \leq 10, \\ 1200 & \text{if } 10 < N \leq 100, \\ 1500 & \text{if } 100 < N \leq 500, \\ 2000 & \text{if } N > 500. \end{cases} \tag{11}$$

This tiered strategy prevents under-querying in dense scenarios (leading to missed detections) and over-querying in sparse scenes (increasing noise and slowing convergence). To enhance query initialization, initial query positions are sampled from high-density regions in D^{pred}, using weighted sampling where the selection probability is proportional to density values. This density-oriented initialization, following principles from recent DETR variants [11], [15], focuses queries on potential object clusters from the outset, improving transformer attention allocation for tiny, dense targets.

The DGQ module synergizes with REM by leveraging rich multi-scale features for accurate density prediction, while dynamic queries feed into the DETR decoder to refine bounding box predictions. Compared to prior density-guided methods [14], [15], integration within the large framework offers superior cross-scale adaptability and noise robustness, as experts can specialize in density estimation under varying conditions. This module addresses key DETR limitations for tiny object detection, including slow convergence on small scales and poor handling of density variations, making it essential for real-world remote sensing applications.

D. Progressive Training Strategy

Training large multi-expert detection models presents significant challenges due to parameter explosion, training instability, and gradient propagation issues inherent in directly combining multiple heterogeneous experts. To address these challenges, a progressive joint training paradigm is adopted through structured expert integration, where expert backbones undergo individual pre-training to establish stable and effective feature representations on independent tasks before integration into the complete framework.

General vision experts receive pre-training on COCO and Objects365 datasets to acquire broad visual feature learning capabilities, while remote sensing-specific experts are pre-trained on DOTA and DIOR datasets to capture domain-specific aerial imagery characteristics. This domain-specialized pre-training ensures that each expert develops strong foundational representations aligned with its designated architectural paradigm and task requirements. The individual pre-training stage proves particularly critical for maintaining expert diversity, as it prevents premature feature homogenization that commonly occurs when heterogeneous architectures are trained jointly from initialization.

Expert integration proceeds through gradual introduction in a structured order, beginning with smaller expert combinations and progressively incorporating additional experts while conducting parameter alignment and collaborative optimization at each integration stage [51]. This gradual integration strategy alleviates convergence difficulties common to large expert systems while promoting complementary feature learning among experts with distinct architectural inductive biases. During integration, expert backbone parameters remain frozen to preserve their specialized feature extraction capabilities, while the routing network, gating mechanisms, and detection heads undergo joint optimization. This selective parameter updating strategy significantly reduces GPU memory requirements by eliminating gradient computation for expert backbones, enabling efficient training despite the large model capacity.

The complete detection system emerges through the selection of the best-performing expert combinations based on validation performance across multiple criteria, including overall detection accuracy, scale-specific AP metrics, and cross-domain generalization capability. The entire framework, encompassing REM and DGQ modules alongside selected expert ensembles, undergoes joint end-to-end fine-tuning on target remote sensing datasets to optimize collaborative feature fusion and query allocation strategies. This progressive approach

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Algorithm 1: ScaleBridge-Det: Complete Detection Pipeline

Input: Image $\mathbf{I} \in \mathbb{R}^{B\times 3\times H\times W}$
Output: Detections $\mathcal{R} = \{(\text{bbox}_i, \text{score}_i, \text{class}_i)\}$
// Stage 1: Adaptive Expert Routing
$M \leftarrow \text{BaseBackbone}(\mathbf{I})$; // Extract base features
$[s_1, s_2, \ldots, s_E] \leftarrow \text{Softmax}(\text{FC}(\text{Flatten}(\text{GAP}(M))))$
$\mathcal{A} \leftarrow \{e \mid e \in \text{Top-1}(\mathcal{C}_e)\}$; // Select top-1 per category
// Stage 2: Expert Feature Extraction
foreach *expert* $e \in \mathcal{A}$ **do**
  $F_e \leftarrow \text{Expert}_e(\mathbf{I})$ with frozen parameters;
  $\{F_e^1, F_e^2, \ldots, F_e^L\} \leftarrow \text{FPN/SFP}(F_e)$
end
// Stage 3: Scale-Adaptive Feature Fusion
foreach *pyramid level* $l \in \{1, \ldots, L\}$ **do**
  $\widetilde{F}_e^l \leftarrow \alpha_e \cdot F_e^l$ for each $e \in \mathcal{A}$;
  $g_l \leftarrow \text{GAP}(\text{Concat}(\{\widetilde{F}_e^l \mid e \in \mathcal{A}\}))$;
  $w_l \leftarrow \text{Softmax}(W_l \cdot g_l + b_l)$;
  $F_l^{\text{fused}} \leftarrow \sum_{e\in\mathcal{A}} s_e \cdot w_{l,e} \cdot \widetilde{F}_e^l$;
end
// Multi-head attention for final features
$F^{\text{final}} \leftarrow \text{MultiHead}(F^{\text{fused}}) = \text{Softmax}(\frac{QK^T}{\sqrt{d}})V$;
// Stage 4: Density-Guided Dynamic Query
$D^{pred} \leftarrow \text{DensityHead}(F^{\text{final}})$; // Predict density map
$N \leftarrow \text{round}(\sum D^{pred})$; // Estimate object count
$N_q \leftarrow \begin{cases} 900 & \text{if } N \le 10 \\ 1200 & \text{if } 10 < N \le 100 \\ 1500 & \text{if } 100 < N \le 500 \\ 2000 & \text{if } N > 500 \end{cases}$; // Adaptive query allocation
$Q_{\text{init}} \leftarrow \text{WeightedSample}(D^{pred}, N_q)$; // Sample from density map
// Stage 5: DETR Decoder with Dynamic Queries
$Q \leftarrow Q_{\text{init}}$;
for *decoder layer* $i \leftarrow 1$ **to** N_{layers} **do**
  $Q \leftarrow \text{SelfAttention}(Q)$;
  $Q \leftarrow \text{CrossAttention}(Q, F^{\text{final}})$;
  $Q \leftarrow \text{FFN}(Q)$;
end
// Stage 6: Prediction Heads
$\{\text{bbox}, \text{class}\} \leftarrow \text{PredictionHeads}(Q)$;
$\mathcal{R} \leftarrow \{(\text{bbox}_i, \text{score}_i, \text{class}_i) \mid \text{score}_i > \tau_{\text{conf}}\}$;
return $\mathcal{R}$

substantially reduces training instability while maintaining the representational benefits of diverse expert architectures, providing a robust foundation for Large detection model construction. To balance model performance with computational efficiency during deployment, only the most effective expert combinations identified through validation are retained in the final model architecture.

IV. Experiments

In this section, we first describe the datasets and evaluation metrics used in our experiments. Subsequently, we provide implementation details of the proposed method. Then, we compare ScaleBridge-Det with state-of-the-art approaches on multiple benchmarks. Finally, we validate the effectiveness of key components through comprehensive ablation studies.

A. Datasets and Evaluation Metrics

AI-TOD Dataset: The AI-TODv2 dataset comprises 28,036 aerial images containing 700,621 object instances, with 86% of objects smaller than 16 pixels and an average object size of only 12.8 pixels [52]. This extreme tiny-object emphasis makes AI-TOD a particularly challenging benchmark for evaluating balanced detection performance across scales. The dataset uses a standard split of 11,414 images for training, 2,804 for validation, and 14,018 for testing.

VisDrone Dataset: VisDrone contains 10,209 UAV-captured images with 2000×1500 pixel resolution, covering 10 object categories with splits of 6,471 training, 548 validation, and 3,190 test images [53]. For cross-domain evaluation, we establish a category mapping protocol to align semantically compatible classes between AI-TOD and VisDrone. Specifically, AI-TOD *vehicle* (category 5) is mapped to VisDrone vehicle-related categories {*car, van, truck, tricycle, awning-tricycle, bus, motor*}, and AI-TOD *person* (category 6) is mapped to VisDrone human-related categories {*pedestrian, people*}. Other AI-TOD categories (airplane, bridge, storage-tank, ship, swimming-pool, wind-mill) have no correspondence in VisDrone and are excluded from cross-domain evaluation. Models trained on AI-TOD are directly evaluated on the VisDrone validation set without fine-tuning, providing a rigorous test of cross-domain generalization under distribution shifts between satellite and UAV imaging platforms.

DTOD Dataset: The DTOD dataset presents an orthogonal challenge through extreme object density, with individual images containing over 2,000 object instances on average [54]. This dataset complements our evaluation by testing detection performance in scenarios with overwhelming object crowding, where density-guided query allocation becomes particularly critical for maintaining computational efficiency and detection accuracy.

Evaluation Metrics: We evaluate model performance using standard metrics including Average Precision (AP), AP50 (IoU threshold 0.5), and scale-specific metrics: APvt (very tiny), APt (tiny), APs (small), and APm (medium). For cross-domain evaluation, we report mAP and mAP50 to assess generalization capability.

B. Implementation Details

Our implementation builds upon the MMDetection framework. All models are trained with uniform input resolution of 1024×1024 across all datasets (AI-TOD, VisDrone, and DTOD), using AdamW optimization with a learning rate of 0.0001 and a batch size of 1.

The multi-expert architecture requires a carefully structured training strategy to ensure stable optimization and effective expert collaboration. We employ a three-stage progressive training paradigm: **Stage 1 - Backbone initialization.** Each expert backbone is initialized from its architecture-specific pre-trained weights: ResNet-50 from ImageNet, ViT-L from MAE self-supervised pre-training, and Swin-L from ImageNet-22K

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supervised pre-training. **Stage 2 - Domain-specialized fine-tuning.** Each backbone is then fine-tuned on its designated corpus to develop complementary feature biases. General-vision experts are fine-tuned on COCO/Objects365 to acquire broad visual feature learning capabilities, while remote sensing-specific experts are fine-tuned on DOTA/DIOR to capture domain-specific aerial imagery characteristics. Concretely, ResNet-50 is fine-tuned on COCO, ViT-L on COCO and DOTA, and Swin-L on Objects365 and DIOR. **Stage 3 - Gradual integration and joint optimization.** Backbones are frozen, and only the REM routing network, DGQ module, and detection head are trained. Experts are progressively combined in structured order, beginning with smaller combinations (e.g., ResNet+ViT) and gradually incorporating additional experts (e.g., +Swin), with parameter alignment at each step. The complete framework then undergoes joint fine-tuning on the target dataset. This progressive approach is essential because direct joint training from initialization leads to catastrophic failure (28.3% AP vs. 35.7% AP with progressive training) due to gradient magnitude mismatch, feature homogenization, and routing network confusion when expert capabilities are simultaneously evolving. To balance model performance and computational efficiency, only the most effective expert combinations are retained in the final deployment model.

To preserve pre-trained knowledge and ensure training stability, we adopt a frozen expert strategy where expert backbone parameters remain fixed during joint training, allowing only the routing network and fusion modules to be optimized. Although our REM module incorporates learnable calibration factors (Equation 3) to address feature magnitude imbalances across heterogeneous architectures, these factors are designed for forward-pass feature fusion rather than backward-pass gradient stabilization. Empirical experiments show that unfreezing multiple heterogeneous experts introduces training instability from three compounding factors: gradient magnitude mismatch across CNN and Transformer architectures, bidirectional coupling between routing decisions and evolving expert parameters, and catastrophic forgetting of diverse pre-training knowledge from multiple source domains. The frozen expert strategy achieves superior performance (35.7% AP) compared to partial fine-tuning (34.8% AP) while reducing memory consumption by 18%, validating our architectural design choice.

All methods are benchmarked under identical conditions (single NVIDIA A100 GPU, batch size 1, input resolution 800×800). Our full 3.0B-parameter model requires approximately 1850G FLOPs and achieves an inference speed of 2.32 FPS, compared to DQ-DETR (63M parameters, 245G FLOPs, 18.2 FPS) and CoDETR (64M parameters, 517G FLOPs, 9.75 FPS). While the computational overhead is higher, ScaleBridge-Det brings substantial performance improvements (+5.2% AP over DQ-DETR, +7.3% AP over CoDETR), making it well-suited for precision-oriented offline remote sensing applications such as disaster assessment and precision agriculture, where detection accuracy requirements far exceed real-time constraints.

The primary computational bottleneck arises from the multi-expert feature fusion in REM, which accounts for approximately 46% of the total FLOPs. This is mainly due to the need to dynamically process and integrate features from multiple heterogeneous backbones across different pyramid levels. In addition, the DGQ module's tiered query allocation is designed to prevent Hungarian matching's quadratic complexity $\mathcal{O}(N_q^2 \cdot N_{gt})$ from causing OOM errors in extremely dense scenes while maintaining sufficient query coverage for recall.

The proposed large-scale multi-expert architecture is primarily designed for precision-oriented offline scenarios rather than real-time deployment scenarios. Typical use cases include satellite image analysis for disaster assessment and large-scale geographic surveying, where images are commonly processed in batch mode and inference latency is not a critical constraint. For resource-constrained or latency-sensitive applications, such as onboard UAV deployment, a compact detector distilled from ScaleBridge-Det can serve as a more practical alternative, enabling improved deployment efficiency while retaining the knowledge learned by the teacher model.

From an implementation and reproducibility perspective, the proposed framework is built upon the publicly available MMDetection codebase, with all expert backbones initialized using publicly released pre-trained weights. The three-stage progressive training paradigm follows a well-defined protocol described aforementioned and does not require specialized hardware beyond standard GPU clusters. Moreover, since expert backbones are frozen during joint training, the computational cost of the final fine-tuning stage is substantially reduced, making training feasible on a single A100 GPU.

In terms of performance-overhead trade-off, the proposed method achieves 7.3% AP and 5.2% AP improvements over CoDETR and DQ-DETR, respectively, representing substantial improvements on a challenge task where even 1% AP improvement is considered significant. The frozen-expert strategy further mitigates the overhead by eliminating gradient computation for expert backbones, reducing memory consumption by 18% compared to partial fine-tuning while achieving superior accuracy. This trade-off is therefore favorable for the target application domain, where detection accuracy directly impacts downstream decision-making quality.

C. Comparisons with State-of-the-Art Methods

We compare ScaleBridge-Det with state-of-the-art methods across three challenging benchmarks to demonstrate its effectiveness and generalization capability. The comprehensive comparison includes both general object detection methods and specialized remote sensing approaches. Figure 9 presents per-category precision-recall curves on AI-TOD-v2 test set, demonstrating consistent performance improvements across all object categories. Figure 10 shows per-scale precision-recall curves, validating balanced detection performance from very tiny to large objects.

Performance on DTOD Dataset. To validate the effectiveness on extremely dense tiny object scenarios, we evaluate on the DTOD dataset. Table I presents the comparison with state-of-the-art methods. ScaleBridge-Det achieves 29.8% AP50, surpassing the previous best SCDNet (24.2% AP50) by 5.6 percentage points. The superior performance demonstrates our

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Method	Venue	Backbone	Param	AP_{50}	AP_{75}	AP^S_{50}	AP^M_{50}	AP^L_{50}
Faster R-CNN [21]	NeurIPS'15	ResNet-50	41M	11.6	2.1	14.8	9.0	22.5
RetinaNet [22]	ICCV'17	ResNet-50	31M	5.2	0.7	8.9	4.8	8.7
Cascade R-CNN [55]	CVPR'18	ResNet-50	68M	9.9	1.7	12.9	8.0	20.7
YOLOv3 [56]	2018	DarkNet-53	61M	4.4	0.1	5.6	4.2	14.1
FCOS [57]	ICCV'19	ResNet-50	31M	7.0	1.2	9.3	5.6	10.7
RepPoints [58]	ICCV'19	ResNet-50	31M	5.0	0.8	7.2	5.1	10.8
TridentNet [59]	ICCV'19	ResNet-50	32M	2.3	1.0	3.3	3.4	10.2
ATSS [60]	CVPR'20	ResNet-50	31M	14.6	2.2	14.0	11.9	29.7
YOLOv5 [61]	2021	CSPDarkNet-s	26M	13.8	2.4	17.6	10.3	29.2
FoveaBox [62]	TIP'20	ResNet-50	36M	12.0	1.8	12.7	10.2	28.0
TPH-YOLOv5 [34]	ICCV'21	CSPDarkNet-s	27M	10.1	1.9	14.8	7.0	24.8
Dyhead [63]	CVPR'21	ResNet-50	38M	10.5	1.6	12.1	8.2	24.5
YOLOv8 [64]	2023	CSPDarkNet-s	12M	10.0	1.7	12.2	7.5	23.1
RT-DETR [65]	CVPR'24	HGNetV2	31M	7.6	0.6	9.4	5.4	18.7
DDOD [66]	TMM'24	ResNet-50	32M	14.5	2.2	14.9	11.9	28.6
DINO [11]	ICLR'23	ResNet-50	47M	8.5	1.1	12.2	7.0	12.6
KLDNet [67]	TGRS'24	ResNet-50	45M	13.3	2.1	13.4	11.3	27.9
YOLOv10 [68]	NeurIPS'24	CSPDarkNet-s	7M	11.9	2.3	13.6	9.2	20.5
YOLOv11 [69]	2024	C3k2Net-s	8M	12.2	2.2	13.9	9.5	23.8
SCDNet [54]	TGRS'24	CSPDarkNet-s	43M	24.2	3.3	21.0	18.3	43.7
ScaleBridge-Det (Ours)	-	Multi-Expert	2B	**29.8**	**7.2**	**24.0**	**26.0**	**42.0**

TABLE I: Comparison with state-of-the-art object detection methods on the DTOD dataset. AP_{50} denotes the average precision of all categories at IoU = 0.50. AP_{75} denotes the average precision of all categories at IoU = 0.75. AP^S_{50}, AP^M_{50}, and AP^L_{50} denote the average precision of small, medium, and large objects at IoU = 0.50, respectively.

Method	Year	Backbone	Param	AP	AP50	APvt	APt	APs	APm
General Object Detection Methods									
RetinaNet [22]	2017	ResNet-50	33M	8.9	24.2	2.7	8.4	13.1	20.2
Faster-RCNN [21]	2015	ResNet-50	41M	12.8	29.9	0.0	9.2	24.6	37.0
Cascade R-CNN [55]	2018	ResNet-50	70M	15.1	34.2	0.1	11.5	26.7	38.5
DetectoRS [70]	2021	ResNet-50	208M	16.1	35.5	0.1	12.6	28.3	40.0
ATSS [60]	2020	ResNet-50	32M	18.3	43.7	5.2	17.8	25.4	35.1
GFL [71]	2020	ResNet-50	32M	19.5	47.2	6.8	19.1	26.7	36.8
DETR [10]	2020	ResNet-50	41M	18.2	45.3	7.5	17.6	24.8	33.2
Deformable DETR [24]	2021	ResNet-50	40M	22.6	55.1	10.3	22.1	28.5	37.4
DAB-DETR [25]	2022	ResNet-50	44M	23.8	57.6	11.2	23.4	29.8	38.9
DN-DETR [26]	2022	ResNet-50	44M	24.5	59.2	11.8	24.1	30.5	39.6
DINO [11]	2023	ResNet-50	40M	25.9	61.3	12.7	25.3	32.0	39.7
Group DETR [28]	2023	ResNet-50	43M	26.4	62.8	13.1	25.9	32.6	40.3
CoDETR (baseline) [12]	2023	ResNet-50	64M	28.4	66.4	12.8	29.1	30.6	45.4
YOLOv8-M [64]	2023	CSPDarkNet	52M	30.1	62.1	10.6	28.4	40.2	46.5
YOLOv11-M [69]	2024	CSPDarkNet	48M	30.5	63.5	10.3	29.7	40.5	45.5
YOLOv12-M [72]	2025	CSPDarkNet	45M	26.3	56.4	7.9	24.6	35.2	40.5
Remote Sensing Object Detection Methods									
DotD [73]	2021	ResNet-50	41M	20.4	51.4	8.5	21.1	24.6	30.4
NWD-RKA [74]	2022	ResNet-50	68M	22.2	52.5	7.8	21.8	28.0	37.2
RFLA [75]	2022	ResNet-50	26M	25.7	58.9	9.2	25.5	30.2	40.2
QueryDet [33]	2022	ResNet-50	45M	26.8	60.4	12.5	26.3	33.1	41.5
TPH-YOLOv5 [34]	2021	CSPDarkNet	28M	25.3	58.7	11.8	24.9	31.2	39.8
DNTR [35]	2024	ResNet-50	128M	29.0	58.3	17.3	29.2	33.5	40.5
CFENet [9]	2025	PoolFormer	35M	30.2	63.7	12.8	30.3	36.7	41.9
DQ-DETR [14]	2024	ResNet-50	63M	30.5	69.2	15.2	30.9	36.8	45.5
D3Q [15]	2025	ResNet-50	49M	32.1	70.8	16.5	32.5	37.2	46.1
LAE-DINO [46]	2025	Swin-T	180M	27.2	63.8	11.1	26.8	33.9	45.3
DINOv3-7b-sat [16]	2025	ViT-7B	7B	27.4	56.8	9.2	26.5	34.1	45.2
ScaleBridge-Det (ours)	-	Multi-Expert	2B	33.6	71.8	14.2	30.4	35.1	46.8
ScaleBridge-Det (ours)	-	Multi-Expert	3B	**35.7**	**72.1**	**16.2**	**34.5**	**39.5**	**51.4**

TABLE II: Performance comparison on AI-TOD-v2 test set. ScaleBridge-Det achieves state-of-the-art performance with balanced detection across all object scales. Param. denotes model parameters (M: millions, B: billions). Best results are in **bold**.

method's capability in handling dense tiny objects through density-guided dynamic query allocation. Notably, our method maintains balanced performance across different object scales ($AP50^S$: 24.0%, $AP50^M$: 26.0%, $AP50^L$: 42.0%), confirming the generalization of the proposed approach across extreme density scenarios.

Performance on AI-TODv2. Table II presents detailed comparison results on the AI-TODv2 dataset. ScaleBridge-Det with 2.0 billion parameters achieves 33.6% AP, surpassing all baseline methods [8]. The 3.0B parameter variant further improves to 35.7% AP, establishing state-of-the-art performance on AI-TOD. Additionally, our method reaches 71.8% AP50 with the 2.0B model and 72.1% with the 3.0B variant, indicating robust performance across all object scales. The primary

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advancement emerges in handling tiny targets: the 3.0B model achieves 16.2% APvt and 34.5% APt, substantially exceeding the CoDETR baseline's 12.8% APvt and 29.1% APt. This demonstrates the effectiveness of scale-adaptive expert routing and density-guided query allocation for balanced multi-scale detection.

Notably, despite leveraging 7 billion parameters and pre-trained representations from 493M satellite images, DINOv3-7b-sat achieves only 27.4% AP, underperforming several lightweight baselines. This limitation stems from computational constraints that restrict trainable parameters to only the final two transformer blocks, preventing effective feature adaptation for tiny object detection. These findings indicate that large foundation models require sufficient trainable capacity and task-specific architectural design to achieve superior performance in specialized domains.

Cross-Dataset Evaluation on VisDrone with Category Mapping. Table III presents cross-dataset evaluation when training on AI-TOD and testing on VisDrone without fine-tuning. All models use the same top-575 prediction constraint for fair comparison, which is set based on the maximum number of ground-truth objects (573) in a VisDrone validation image, with a small buffer. DQ-DETR completely fails with 2.7% mAP. CoDETR shows poor generalization (ResNet50: 11.8% mAP, Swin-L: 17.6% mAP). Notably, LAE-DINO achieves 29.5% mAP and 57.5% mAP50, demonstrating comparable robustness to ScaleBridge-Det despite moderate single-dataset performance (27.2% AP in Table II). This robustness benefits from large-scale pre-training on diverse remote sensing data (LAE-1M dataset with 1.2M images). ScaleBridge-Det achieves the best performance with 30.7% mAP and 58.8% mAP50, demonstrating effective generalization across different datasets. Visual comparison in Figure 4 further illustrates these differences with predominantly high-quality detections.

Method	Backbone	mAP	mAP50
DQ-DETR	ResNet50	2.7	8.5
CoDETR	ResNet50	11.8	24.6
CoDETR	Swin-L	17.6	40.0
LAE-DINO	Swin-T+BERT	29.5	57.5
ScaleBridge-Det (ours)	**Multi-Expert**	**30.7**	**58.8**

TABLE III: Cross-dataset evaluation (trained on AI-TOD, tested on VisDrone without fine-tuning). All models use top-575 predictions for fair comparison. Category mapping: AI-TOD *vehicle* → VisDrone {*car, van, truck, tricycle, bus, motor*}; AI-TOD *person* → VisDrone {*pedestrian, people*}. DQ-DETR fails completely. ScaleBridge-Det achieves the best performance.

D. Ablation Study

To investigate the contribution of each module in our proposed architecture, we conduct comprehensive ablation experiments on the AI-TOD validation set. We first analyze parameter efficiency by comparing ScaleBridge-Det variants with different expert configurations against state-of-the-art baselines in Figure 5. Each baseline method is represented by a unique combination of marker shape and color for clear visual identification. The results demonstrate that our multi-expert architecture achieves superior parameter efficiency: with 800M parameters (1+1+1 configuration using ResNet+ViT+Swin experts), ScaleBridge-Det already surpasses most baseline methods. The performance scales consistently with model capacity, exhibiting a smooth growth trajectory from 30.2% AP at 800M parameters to 36.8% AP at 7B parameters, where each expert combination contributes complementary features for balanced multi-scale detection. Although the 7B variant achieves a higher AP, we select the 2B and 3B variants in the main experiments to provide a more reasonable balance among model capacity, detection accuracy, and computational cost. All ScaleBridge-Det variants utilize tiny-object-specific pre-training on DIOR and DOTA datasets, which proves essential for capturing fine-grained features characteristic of remote sensing imagery. Notably, at the same 7B parameter scale, our method substantially outperforms the foundation model DINOv3-7B-sat (36.8% vs 27.4% AP, a 9.4 percentage point

REM	DGQ	AP	AP_{vt}	AP_t	AP_s	AP_m
		31.7	13.9	31.3	38.3	50.2
✓		34.2	14.3	32.7	43.6	57.9
	✓	32.6	15.7	33.1	37.3	49.8
✓	✓	**36.7**	**15.8**	**35.2**	**46.3**	**58.3**

TABLE IV: Ablation Study: Component Contributions on AI-TOD-v2 validation set using 2.0B base model.

σ	AP	AP_{vt}	AP_t	AP_s	AP_m
1.0	35.4	**16.5**	34.1	39.0	50.8
1.5	**35.7**	16.2	**34.5**	39.5	**51.4**
2.0	35.5	15.8	34.2	**39.8**	51.0

TABLE V: Sensitivity Analysis: Impact of Gaussian kernel parameter σ on density map construction (AI-TOD-v2 test set, 3×3 kernel, 3.0B model).

Method	Params	AP	AP_{50}	AP_{vt}	AP_t	AP_s	AP_m
DQ-DETR (original)	63M	30.5	69.2	15.2	30.9	36.8	45.5
DQ-DETR-MoE	3.0B	32.2	71.3	14.8	30.3	36.3	49.6
ScaleBridge-Det (ours)	3.0B	**35.7**	**72.1**	**16.2**	**34.5**	**39.5**	**51.4**

TABLE VI: Fair comparison with equivalent parameters on AI-TOD-v2 test set.

Method	Routing Design	AP	AP_{vt}	AP_t	AP_s	AP_m
Standard MoE (baseline)	Top-k routing	32.5	13.0	29.6	34.0	44.6
+ Cat.-wise selection	Scale-aware routing	32.6	13.8	29.5	34.4	45.2
+ Calibration	+ Expert calibration	32.8	13.6	29.8	34.5	45.8
REM (full, ours)	+ Dual-weight fusion	**33.6**	**14.2**	**30.4**	**35.1**	**46.8**

TABLE VII: REM routing design ablation (without DGQ module) within the full multi-expert framework on AI-TOD-v2 test set.

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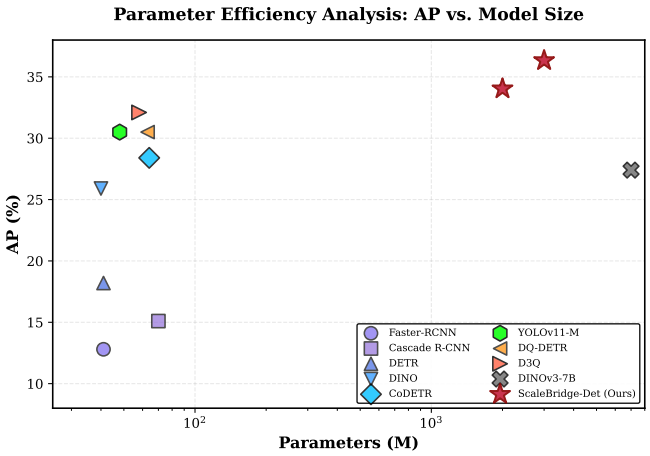


Fig. 5: Parameter Efficiency Analysis: AP vs. Model Size on AI-TOD test set. Each baseline method is represented by a unique marker shape and color. The red star line shows ScaleBridge-Det variants with different expert configurations.

improvement), demonstrating that task-specific multi-expert architectural design with progressive training surpasses generic foundation model pre-training for specialized tiny object detection tasks.

Homogeneous vs. Heterogeneous Expert Configurations. To validate the necessity of architectural diversity, we compare homogeneous configurations (3×ResNet-50, 3×ViT-L, 3×Swin-L) with our heterogeneous design (ResNet-50+ViT-L+Swin-L) using 3 experts in all configurations. Each expert undergoes pre-training on large-scale general datasets: ResNet-50 is initialized from ImageNet and fine-tuned on COCO; ViT-L is initialized from MAE pre-training and fine-tuned on COCO and DOTA; Swin-L is initialized from ImageNet-22K pre-training and fine-tuned on Objects365 and DIOR. As shown in Table VIII, ResNet excels at extremely tiny objects (APvt=18.3%) through strong local receptive fields, ViT achieves the best overall AP (32.9%) and medium object performance (APm=51.6%) via global attention, while Swin demonstrates balanced performance across scales. Our heterogeneous combination surpasses all homogeneous configurations (AP=33.2, +0.3 over best homogeneous), proving that performance gains come from complementary architectural diversity rather than increased expert count.

Expert Configuration	#Experts	AP	AP_{vt}	AP_t	AP_s	AP_m
3×ResNet-50 (homogeneous)	3	30.7 ±0.1	**17.8** ±0.46	31.0 ±0.1	33.8 ±0.06	44.2±0.06
3×ViT-L (homogeneous)	3	33.1 ±0.18	12.1 ±0.38	32.4 ±0.1	35.4 ±0.15	51.4±0.2
3×Swin-L (homogeneous)	3	32.8 ±0.15	15.4 ±0.2	33.8 ±0.06	39.6 ±0.25	51.2 ±0.3
ResNet-50+ViT-L+Swin-L (ours)	3	**34.1** ±0.06	16.7 ±0.25	**34.8** ±0.06	**39.9** ±0.21	**51.9** ±0.21

TABLE VIII: Homogeneous vs. heterogeneous expert configurations on AI-TOD-v2 test set with the same expert numbers.

Table IV shows the performance under different combinations of the Routing-Enhanced Mixture Attention (REM) module and Density-Guided Dynamic Query (DGQ) module.

Component Contributions. The REM module independently contributes a notable 2.5 percentage point AP improvement over the baseline (from 31.7% to 34.2%), demonstrating the substantial effectiveness of scale-adaptive expert routing and multi-scale feature fusion. The DGQ module shows particular strength in improving very tiny object detection, with a 1.8 percentage point APvt improvement (from 13.9% to 15.7%). The full system combining both modules achieves 36.7% AP with 15.8% APvt, representing a 5.0 percentage point overall gain and 1.9 percentage point improvement on very tiny targets.

Synergistic Effects. The combined model achieves the best overall performance, with improvements beyond those obtained using REM or DGQ alone. These results suggest that scale-adaptive feature fusion and density-guided query allocation provide complementary benefits. The proposed method not only enhances tiny-object detection but also promotes more balanced performance across different object scales.

Implementation Details. For component ablation, we use standard FPN routing as the non-REM baseline. For DGQ ablation, we employ fixed query mechanisms with 900 object queries, eliminating density map generation and dynamic allocation. This systematic approach ensures accurate attribution of performance improvements to specific architectural components.

Hyperparameter Sensitivity Analysis. Table V evaluates the impact of Gaussian kernel parameter σ on density map construction. Overall AP remains highly stable (35.4–35.7%, variation <0.3%), demonstrating robustness to hyperparameter choices. We select σ=1.5 as it achieves the best overall AP (35.7%) and balanced performance across scales.

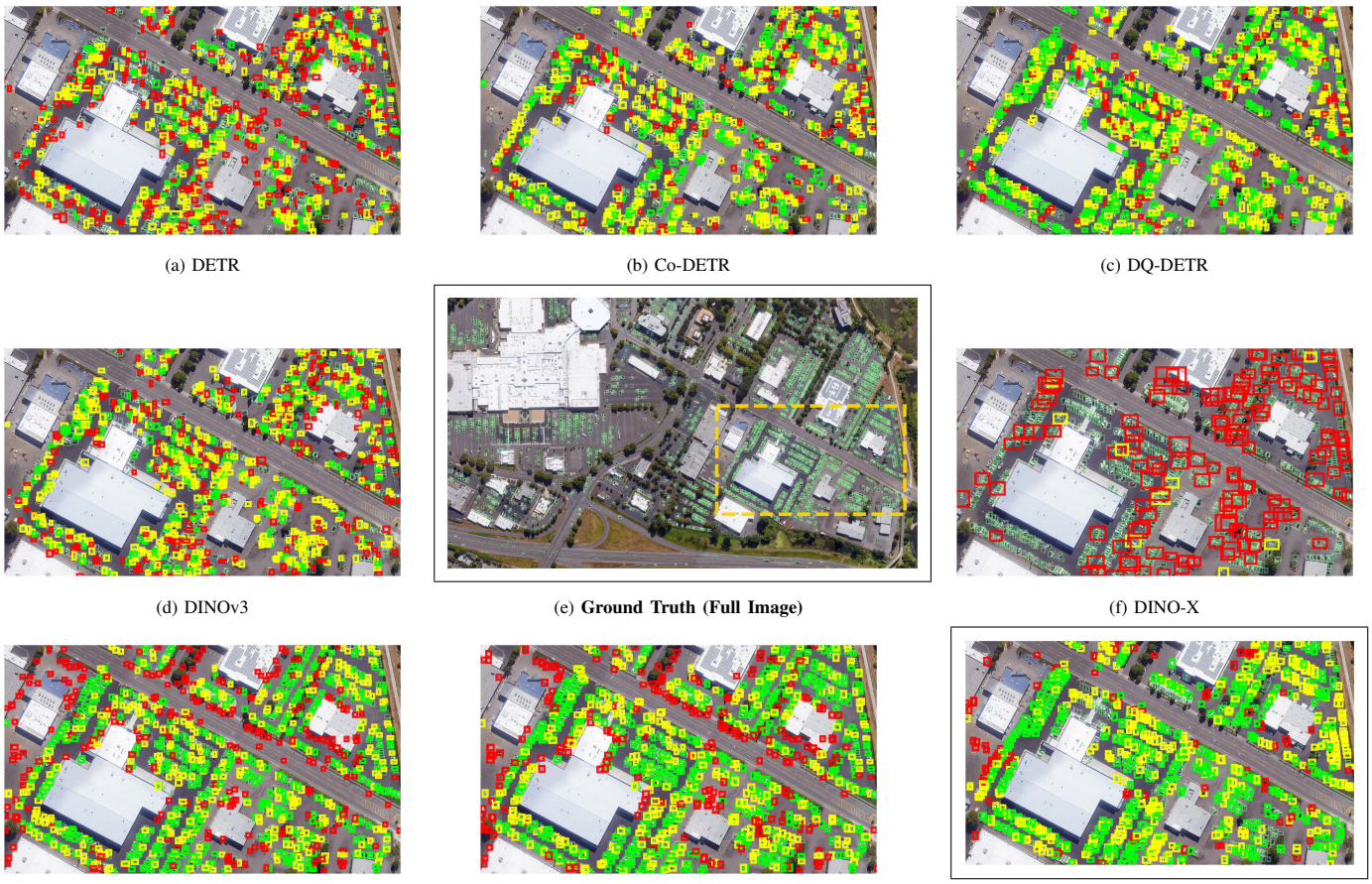
Fair Comparison with Equivalent Parameters. To ensure fair comparison, Table VI presents DQ-DETR-MoE, which extends DQ-DETR with multi-expert architecture while maintaining the original density query mechanism. With 3.0B parameters, DQ-DETR-MoE achieves 32.2% AP, demonstrating that simply scaling up parameters provides limited gains (+1.7% over original 63M DQ-DETR). In contrast, ScaleBridge-Det achieves 35.7% AP with the same 3.0B parameters, representing a +3.5% improvement over DQ-DETR-MoE. This substantial gap validates that our innovations (REM's scale-adaptive routing and DGQ's density-guided allocation) are the key drivers of performance gains, rather than merely increased model capacity.

REM Design Ablation. Table VII further isolates the contribution of each innovation within the REM module, using Standard MoE routing as the baseline within the full multi-expert framework. The category-wise expert selection provides the largest APvt gain (+0.8 pp, from 13.0% to 13.8%), confirming that balanced activation across scale categories benefits tiny-object detection. Adding expert-specific calibration factors improves overall AP (+0.2 pp) and medium-object APm (+0.6 pp, from 45.2% to 45.8%), addressing the magnitude imbalance between CNN (ResNet) and Transformer (ViT, Swin) architectures. The dual-weight fusion mechanism provides the largest single-step AP gain (+0.8 pp) and further enhances all metrics. The full REM achieves 33.6% AP and 14.2% APvt, representing +1.1% and +1.2% improvements over the Standard MoE baseline, respectively.

Routing Behavior Analysis. To validate REM's cross-domain adaptability, we analyze expert activation patterns on both in-domain (AI-TOD, 2,804 images) and cross-domain

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(a) DETR (b) Co-DETR (c) DQ-DETR

(d) DINOv3 (e) **Ground Truth (Full Image)** (f) DINO-X

(g) YOLOv8 (h) YOLOv11 (i) **ScaleBridge-Det (Ours)**

Fig. 6: Detection comparison on an extremely dense scene of DTOD dataset (3058 objects). Center (e): **Ground Truth** with golden dashed box marking the zoom region. Surrounding subfigures show zoomed detection results from different methods. Light green boxes: ground truth; colored predictions: green (IoU$\geq$0.5), yellow (0.2$\leq$IoU$<$0.5), red (IoU$<$0.2).

(VisDrone, 548 images) datasets. The category-wise top-1 selection guarantees balanced activation across expert categories by design. As shown in Figure 7, ResNet concentrates activation on tiny objects, ViT on large structures, and MoE adaptively fuses complementary features. Figure 8 quantitatively confirms this specialization: ResNet achieves the highest activation contrast for Very Tiny and Tiny objects, while ViT excels on larger scales, and MoE maintains strong performance across all scales through adaptive fusion. This consistent specialization across domains demonstrates that REM genuinely adapts routing rather than merely relying on pre-training.

Progressive Training and Expert Freezing Analysis. Table IX compares three training strategies on AI-TOD-v2 test set. Direct joint training fails catastrophically (28.3% AP), validating the necessity of progressive training. The frozen expert strategy achieves superior performance (35.7% AP, 16.2% APvt) compared to partial fine-tuning (34.8% AP, 15.4% APvt) while reducing memory consumption by 18% (78GB vs. 96GB). A detailed analysis of the underlying instability factors is provided in Section IV-B.

Strategy	AP	APvt	APt	Memory	Time
Direct joint training	28.3	12.1	27.5	112GB	1.5×
Progressive + frozen (ours)	**35.7**	**16.2**	**34.5**	78GB	1.0×
Progressive + partial fine-tuning	34.8	15.4	33.6	96GB	1.2×

TABLE IX: Impact of training strategies on AI-TOD-v2 test set using 3.0B model.

E. Visualization Analysis

Figure 6 presents qualitative analysis on an extremely dense DTOD scene containing 3058 objects, with a zoom region of 890 densely packed tiny objects. DINO-X, despite strong performance on general benchmarks, exhibits catastrophic failure with only 108 detections (12% recall), confirming the domain gap between general and remote sensing detection. CoDETR and DQ-DETR show moderate improvements but still suffer from substantial missed detections. ScaleBridge-Det achieves 659 detections (74% recall) with predominantly high-quality predictions (IoU$\geq$0.5), representing a 6× improvement over DINO-X, validating the effectiveness of density-guided query allocation combined with scale-adaptive expert routing.

V. Conclusion

In this work, we propose ScaleBridge-Det, a large detection framework designed for tiny objects with scale-adaptive expert routing and density-guided query allocation to achieve

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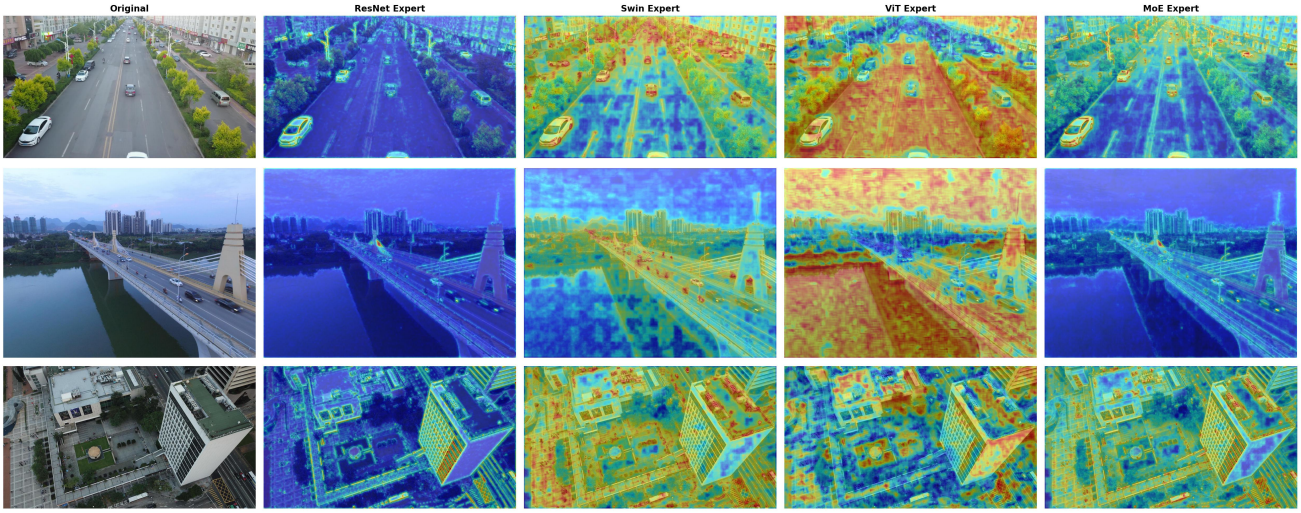


Fig. 7: Expert activation heatmap comparison across different backbone types on VisDrone test scenes. Warmer colors indicate stronger activation.

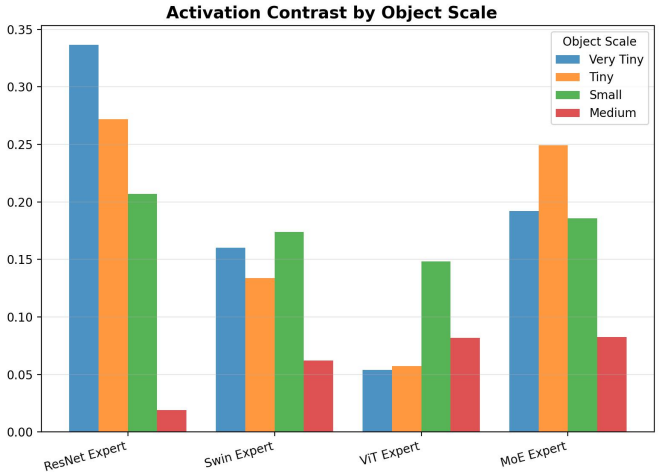


Fig. 8: Backbone activation contrast analysis by object scale.

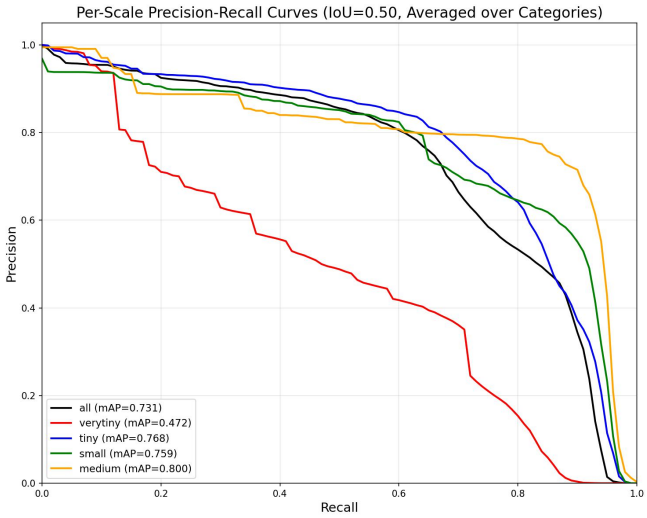


Fig. 10: Per-scale precision-recall curves on AI-TOD-v2 test set.

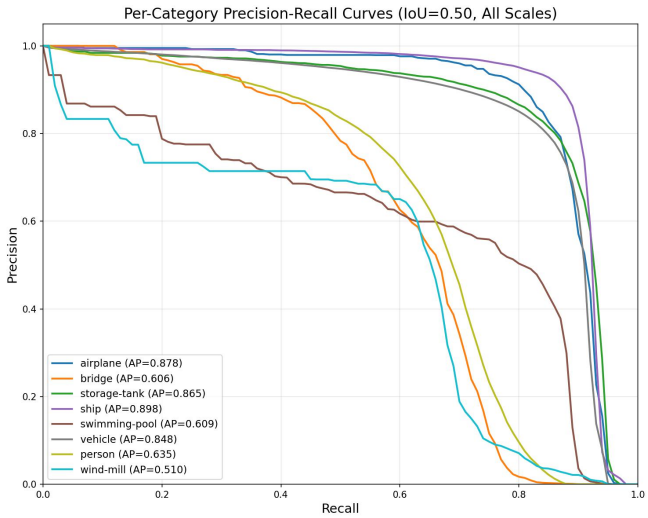


Fig. 9: Per-category precision-recall curves on AI-TOD-v2 test set.

balanced performance across diverse scales. ScaleBridge-Det incorporates a Routing-Enhanced Mixture Attention module that leverages a mixture-of-experts architecture with adaptive routing mechanisms to dynamically integrate scale-specific expert features, effectively resolving the scale competition problem where tiny objects are suppressed by dominant large structures. Additionally, the Density-Guided Dynamic Query module provides spatial awareness of object distribution patterns and adaptively adjusts query allocation according to predicted density maps, enabling efficient resource distribution across varying density scenarios while maintaining computational efficiency. Extensive experiments on challenging benchmarks demonstrate that the framework achieves balanced detection performance across all object scales, from very tiny targets occupying fewer than eight pixels to large structures spanning hundreds of pixels, while exhibiting superior cross-domain generalization capability. The final fine-tuning stage of the complete framework can be reproduced on a single A100 GPU following the provided training setting. Despite these advantages, the multi-expert design inevitably introduces

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additional computational overhead, which may limit its deployment in real-time scenarios. Nevertheless, the favorable accuracy–efficiency tradeoff makes the framework particularly suitable for precision-oriented offline applications, such as disaster assessment and precision agriculture, where detection accuracy is prioritized over inference speed. Future research will focus on developing lightweight variants suitable for resource-constrained UAV platforms, for example by leveraging ScaleBridge-Det as a teacher for knowledge distillation into compact detectors, or by exploring dynamic sparse expert activation to reduce inference cost. We also plan to extend the framework to accommodate additional remote sensing modalities, such as synthetic aperture radar and hyperspectral imagery.

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